Clinical Co-Pilot — Complete Project Walkthrough (CIOS)

Clinical Co-Pilot is an AI-powered **Clinical Intelligence Operating System (CIOS)** that transforms real-time patient-doctor conversations into structured medical records, financial claims, and longitudinal outcome data.

🏗️ Architecture Overview

```
graph TD
    User((Therapist)) -->|Ambient Audio| DG(Deepgram Nova-2-Medical)
    DG -->|Live Transcript| UI[Next.js Dashboard]
    UI -->|Full Context| GPT(OpenAI GPT-4o)
    GPT -->|SOAP/DAP/GIRP| UI
    GPT -->|ICD-10/CPT Suggestions| UI
    GPT -->|Risk Assessment| UI
    UI -->|Persist| DB[(Supabase PostgreSQL)]
    DB -->|RLS Protected| PP[Patient Portal]
```

Data Flow

1. **Therapist clicks "Start Session"**: Audio streams to Deepgram (WebSocket).
2. **Real-time Diarization**: Speaker 0/1 identified and displayed in chat bubbles.
3. **Session Stop**: Full transcript + duration sent to GPT-4o.
4. **Multi-Format Synthesis**: AI generates SOAP, DAP, or GIRP notes plus Risk Score.
5. **Invisible Billing**: AI extracts symptoms to map ICD-10 codes and suggests CPT codes.
6. **Persistence**: Transcript, Notes, Billing, and Risk data saved with RLS protection.

🚀 Key Modules & UI Features

1. The Clinical Scribe (Ambient Dashboard)

- **LiveTranscript**: Chat-style interface with auto-scroll and speaker labels.
- **Intelligence Panel**: Dynamically switches between AI drafting and Entity extraction.
- **Note Formats**: User-selectable documentation styles (SOAP, DAP, GIRP).

2. Billing & Revenue Ops (Superbill Generator)

- **Automatic Coding**: Maps extracted symptoms to ICD-10 diagnosis codes.
- **Fee Management**: Auto-suggests CPT codes based on session minutes (e.g., 90834).
- **Billing History**: Tracks claim status (Draft, Ready for Payer, Finalized).

3. Outcome Intelligence (PHQ-9/GAD-7)

- **Integrated Assessments**: Direct clinical assessment gathering within the workflow.
- **Clinical Flywheel**: Uses assessment scores to enrich the clinical documentation.

4. Appointment Management

- **Inbox**: Real-time tracking of new appointment requests.
- **Lifecycle**: Pending → Scheduled → Completed transitions.

🛠️ Technical Stack

Layer	Technology
Core Framework	Next.js 15 (TypeScript, App Router)
Styling	TailwindCSS + Lucide Icons
Speech-to-Text	Deepgram (Medical Model)
Intelligence Engine	OpenAI GPT-4o (Structured Mode)
Database/Auth	Supabase (PostgreSQL, Auth, Storage)
Middleware	Edge-side Role Based Access Control (RBAC)

📂 Project Structure

```
clinical-copilot/
├── src/app/
│   ├── (auth)/            # Login/Signup/Role Selection
│   ├── doctor/            # Clinician Dashboards & Billing
│   ├── patient/           # Patient Portals
│   ├── session/           # Active Recording Interface
│   ├── actions.ts         # Centralized Server Actions
│   └── middleware.ts      # Route guards & Role routing
├── src/components/        # Reusable UI (Superbill, Recorder, NoteGenerator)
├── src/lib/               # prompts.ts, supabase-client.ts
└── supabase/              # SQL Schema & Migrations
```

✅ Validations & Security

- **HIPAA Readiness**: Row Level Security (RLS) ensures clinicians only see their own patients.
- **Risk Alert Engine**: Automatically catches "high" or "critical" risk factors in conversations.
- **Clinical Consistency**: Strict JSON mode for GPT-4o ensures valid data schemas.

Generated on February 16, 2026